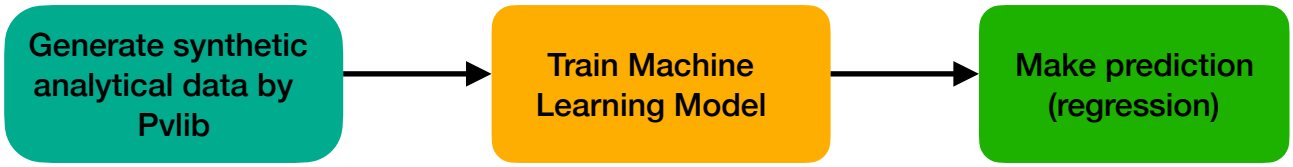
Solar Panel Project

Group 2

General approach:



Our objective is to predict solar panel parameters using power generation data, and we achieve this through a three-step process. First, we use the PVlib library to generate synthetic datasets that simulate solar power output across various locations, panel specifications, and time periods. This allows us to create a diverse set of scenarios reflecting different environmental conditions. Next, the synthetic data is used to train a machine learning model. During this phase, we perform data analysis and feature engineering to extract key information relevant to predicting panel parameters. Features such as irradiance levels, temperature coefficients, and historical performance metrics are engineered to improve model accuracy. Finally, once the model has been trained, it will be applied to predict solar panel parameters using real-world power generation data. The model's predictions are validated against actual panel performance data, ensuring its effectiveness in practical applications.

Through these steps—synthetic data generation, machine learning model training with feature engineering, and application to real-world data—we aim to accurately predict solar panel parameters based on observed power generation.

Data analysis & Feature Engineering

A thorough data analysis and feature engineering process has been conducted to extract more valuable insights for training an effective neural network model. Our analysis revealed that the panel orientation angle significantly influences the time at which maximum power generation is reached within a single day. In contrast, the tilt angle of the solar panels exerts a considerable impact on annual power output, meaning that adjustments to the tilt angle can alter the total yearly power production. For simplicity, we assume that the capacity of the solar panel has a linear scaling effect on power generation. This assumption facilitates the modeling process by reducing complexity without sacrificing predictive accuracy.

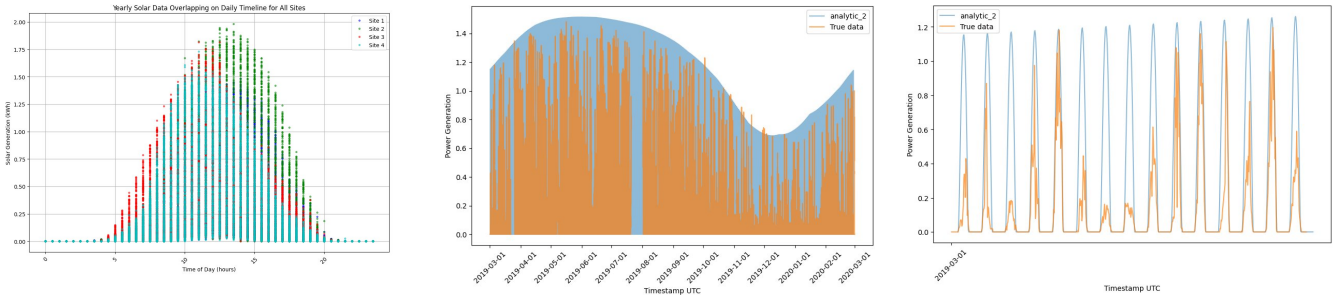


Figure 1: (Left) A scatter plot showing full-year data collected from four real sites plotted over a single day timeframe. The figure illustrates that daily power generation follows a bell-shaped curve, with the panel's orientation angle determining the time at which peak power is achieved. (Middle) A comparison between analytical data generated using PVlib and actual case data on an annual scale. By modifying the input parameters, it was found that the annual power generation curve correlates strongly with the tilt angle. (Right) A comparative analysis of analytical data versus real-world data on a weekly basis.

In our feature engineering efforts, we have identified several key factors that could enhance the performance of the neural network. The first factor is the time of day when maximum power is reached, t_m, which aids in predicting the optimal panel orientation.

The second factor is the normalized maximum power difference, denoted as α, which expressed as: $\alpha = \frac{P_{max} - P_{min}}{P_{avg}}$. α represents the fluctuation level of the power generation within a year and is highly depends on the tilt angle. where P_{max}, P_{min}, P_{avg} represent two peak power output, and the average power output over the year. This metric reflects the fluctuation level of power generation throughout the year and is heavily influenced by the tilt angle. To evaluate the relevance of these impact factors to the objective features, we utilized the mutual information method. Currently, we are also exploring additional potential features that may further improve prediction accuracy in future iterations of our model.

Neural Network Design

For our initial approach to addressing this regression problem, we opted for a simple neural network architecture. Utilizing the PyTorch framework, we developed an enhanced neural network model optimized through the incorporation of advanced techniques such as batch normalization and dropout. These methods were employed to bolster the robustness and performance of the neural network. The model was trained exclusively on synthetic analytical data, with final testing conducted using real-world data collected from four distinct sites.

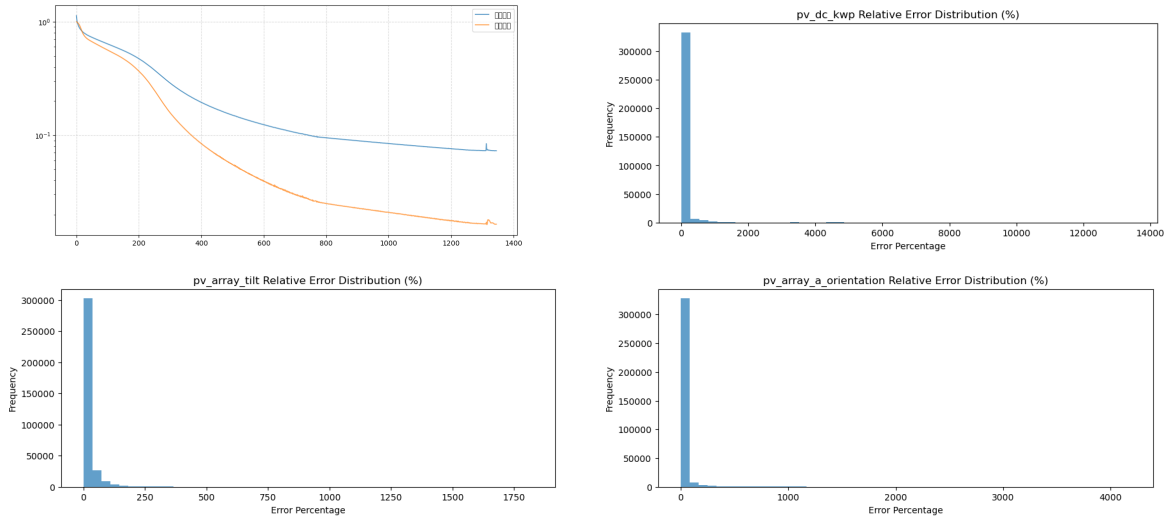


Figure 2: (Upper Left) Convergence of neural network training. (Upper Right) Relative error distribution for pv_dc_kwp. (Lower Left) Relative error distribution for pv_array_tilt. (Lower Right) Relative error distribution for pv_array_a_orientation.

Training concluded after approximately 1,300 steps, at which point the minimum validation loss was achieved. Evaluation with real data revealed the Mean Absolute Percentage Errors (MAPE) for pv_dc_kwp, pv_array_tilt, and pv_array_a_orientation to be 90.27%, 27.32%, and 43.93%, respectively. Analysis of the error distribution histograms indicated that the primary sources of error were due to extreme outliers in a few sample points. These outliers likely contained noise or weather data anomalies, which introduced significant bias into the overall error metrics. To mitigate these issues, further refinement of the model may involve additional preprocessing steps to clean and normalize the input data more effectively. Additionally, exploring alternative architectures or hyperparameter configurations could potentially reduce the impact of outliers and improve overall prediction accuracy. By addressing these challenges, we aim to enhance the reliability and precision of our solar panel performance predictions, contributing to more efficient energy system designs.

Conclusion

In this study, we employed a straightforward neural network approach to address the regression challenges associated with predicting solar panel performance metrics. Utilizing synthetic data for training and real-world data for validation, our model incorporated advanced techniques such as batch normalization and dropout to enhance robustness and predictive accuracy. Despite achieving convergence at approximately 1,300 training steps, our final evaluation using real data indicated varying degrees of success across different metrics. Specifically, we observed Mean Absolute Percentage Errors (MAPE) of 90.27% for pv_dc_kwp, 27.32% for pv_array_tilt, and 43.93% for pv_array_a_orientation. The primary source of error was attributed to outliers in the dataset, which skewed the overall performance metrics. Despite these findings indicating areas for improvement, they also underscore the potential of neural networks in accurately predicting solar energy system performance when provided with additional real-world data and optimized neural network architectures.

Future work

Looking ahead, there are several avenues for further development and refinement. Firstly, continued exploration into the identification and integration of additional impact factors could significantly benefit the training process, leading to more accurate predictions. Secondly, modifying the neural network structure to predict site parameters holistically rather than on a per-sample basis may reduce the influence of outliers and improve overall prediction consistency. Additionally, incorporating more sophisticated preprocessing steps to clean and normalize input data will be crucial in minimizing noise and anomalies that currently bias the predictions. Lastly, experimenting with alternative architectures or hyperparameter settings could provide further enhancements, ultimately contributing to the development of a more reliable and precise predictive model for solar panel performance.

By addressing these future directions, we aim to refine our predictive capabilities, making them more applicable to real-world scenarios and enhancing the prediction performance .